Total No. of Questions : 8]

SEAT No. :

**PD4904**

[Total No. of Pages : 2

**[6404]-462**

**B.E. (Computer Engineering) (Honours)**

**VIRTUAL REALITY AND AUGMENTED REALITY**

**Application Development Using Augmented Reality and Virtual Reality**

**(2019 Pattern) (Semester - VIII) (410703)**

*Time : 2½ Hours]* *[Max. Marks : 70*

*Instructions to the candidates:*

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Make suitable assumptions if necessary.*
4) *Figures to the right indicate full marks.*

***Q1)*** a) What is ARCore? Explain the features of ARCore. **[6]**

b) What is Vuforia? Explain the features of Vuforia. **[6]**

c) Explain anatomy of SLAM algorithm. **[5]**

OR

***Q2)*** a) What considerations should developers keep in mind during the integration of Vuforia into their applications? **[6]**

b) Explain the terms: **[6]**

i) Environmental Detection

ii) Occlusion

c) Give a detailed overview of the key components involved in the structure of a SLAM algorithm. **[5]**

***Q3)*** a) Explain the significance of User Experience (UX) in AR and VR applications, and discuss how programming languages can play a role in improving UX. **[6]**

b) Identify some of the most widely used technologies for Augmented Reality (AR) and Virtual Reality (VR) development. **[6]**

c) Describe the fundamental Object-Oriented Programming (OOP) principles in C# and explain how they are implemented in Unity development. **[6]**

OR

*P.T.O.*

***Q4)*** a) How can Visual Studio be set up for C# development in Unity? **[6]**

b) What is the role of C# in AR and VR development with Unity? **[6]**

c) How does C# manage and control 3D models in Unity? **[6]**

***Q5)*** a) What is the structure and working principle of the HTC Vive VR device? **[6]**

b) Compare the Oculus Rift and HTC Vive in terms of structure, functionality, features, and pros and cons. **[6]**

c) What is the purpose of an AR tracking system in AR technology? **[5]**

OR

***Q6)*** a) What is the role of a tracking system in AR, and how does it enhance the AR experience? **[10]**

b) What are the main advantages and disadvantages of AR and VR, and how do these affect their use across industries? **[7]**

***Q7)*** a) Why are human factors important in designing AR and VR experiences? **[6]**

b) How does the widespread use of AR and VR influence social interactions and societal norms? **[6]**

c) What are the applications of AR and VR in healthcare, and what are the benefits for patients and professionals? **[6]**

OR

***Q8)*** a) How are AR and VR used in science and engineering for simulations, training, and collaborative research? **[6]**

b) What are the applications of AR and VR in the aerospace and defense sectors? **[6]**

c) How do AR and VR impact the gaming and entertainment industry, and what advancements enhance user experiences? **[6]**

× × ×